

MAT9004 Assignment 1

Due: Wednesday, 22 April 2026, 23:55 PM (Week 7)

Please write up your solutions neatly! In particular, use notation and terminology correctly and include some necessary comments. Partial marks will be given for showing that you know some aspects of the answer, even if your solution is incomplete. The questions are NOT arranged in order of difficulty, so you should attempt every question.

Both handwritten and typed solutions are accepted. Please make sure you convert all your work into **on single pdf file** as other types might not display properly on Moodle.



1. [10 points] In a machine-learning system for automated anomaly detection, raw input signals are converted into a single scalar index x , representing a normalized signal intensity. To stabilize training and improve interpretability, the feature undergoes a nonlinear transformation before being passed into a probabilistic classifier.

The feature transformation is defined by

$$f(x) = \ln \left(\frac{x}{1-x} \right).$$

The classifier then maps the transformed feature to a confidence score using

$$g(u) = \frac{1}{1 + e^{-u}}.$$

- (a) [1 point] Determine the maximal domain of f .

- (b) [1 point] Determine the range of f .

The complete transformation used by the system is the composite function

$$h(x) = (g \circ f)(x) = g(f(x))$$

- (c) [1 point] Find an explicit expression for $h(x)$.

- (d) [2 points] Determine the domain and range of h .

For model interpretability, engineers wish to recover the original feature value x from the transformed value $u = f(x)$.

- (e) [3 points] Calculate $f'(x)$ and use it to explain why the inverse of f exists. Find the inverse function $f^{-1}(x)$

- (f) [2 points] State the domain and range of f^{-1} .

2. [8 points] Similar to the area, we can use integration to calculate the volume of the solid formed by rotating a region around the x-axis. If the region to be rotated is bounded by the curve with equation $y = f(x)$, the lines $x = a$, $x = b$ and the x-axis, then

$$V = \int_a^b \pi y^2 dx.$$

Now consider the parabola $f(x) = ax^2 + bx + c$. The curve passes through the origin, remains above the x -axis for all $x \in [0, 1]$. The area enclosed by the parabola together with the x-axis and the line $x = 1$ is $\frac{1}{3}$. Among all parabolas satisfying these conditions, determine the values of a , b and c for which the volume of the solid generated by rotating the curve about the x-axis over the interval $[0, 1]$ is minimized.

3. [15 points] (**Inside trading**) A family decided to invest their \$100,000 savings into the stock market. From a trusted source they know the price fluctuations of an asset shares over the next three years. The price is given by a function

$$f : [0, 3] \rightarrow \mathbb{R},$$

where

$$f(t) = (t - 1)^3(20t^3 - 104t^2 + 143t - 19) + 20.$$

You need to find out how much money the family can raise within the given period (rounded to the nearest integer in thousand dollars) assuming they are not caught by the police.

- (a) [5 points] Compute the derivative $f'(t)$ and find all stationary points of f .
 - (b) [4 points] On which intervals in $t \in [0, 3]$ is $f(t)$ increasing, and where is it decreasing? Classify all stationary points of f as a local minimum, local maximum or neither.
 - (c) [3 points] Find the global minimum and the global maximum of f .
 - (d) [1 point] Assume that the family is cautious so they buy shares only once and sell shares only once. How much money can they make by the end of the third year?
 - (e) [2 points] How much money can they make by buying/selling shares several times?
4. [7 points] An engineering team is analyzing a signal processing system in which four independent input signals, with amplitudes a , b , c and d , are combined through three different measurement channels. Due to the design of the system, each measurement records a linear combination of the four signal amplitudes.

The outputs of the three measurement channels are given by the following equations:

$$\begin{aligned}a + b - 3c - d &= 1 \\3a - b - 3c + 4d &= 4 \\a + 5b - 9c - 8d &= 0\end{aligned}$$

Here, the right-hand sides represent the observed output values from the sensors after calibration. The variables a , b , c and d represent unknown signal strengths at different components of the system.

- (a) [1 point] Write the above system of linear equations in matrix form.
- (b) [6 points] Determine whether the system has a solution, and if so, describe the set of all possible signal configurations consistent with the observed measurements. Express your final answer in matrix form.